Input system:
Dependent variables:x_1
Independent variables:t_1, t_3, t_2

$$\frac{\partial x_1}{\partial t_1} = x_1 \sin(x_1)$$
$$\frac{\partial x_1}{\partial t_2} = x_1$$
$$\frac{\partial x_1}{\partial t_3} = 5$$

==========

Result:
Dependent variables:s_2, x_1, s_1
Independent variables:t_1, t_3, t_2
Additional variables:

$$s_1 = \sin(x_1)$$
$$s_2 = \cos(x_1)$$
$$\frac{\partial x_1}{\partial t_1} = x_1 s_1$$
$$\frac{\partial x_1}{\partial t_2} = x_1$$
$$\frac{\partial x_1}{\partial t_3} = 5$$
$$\frac{\partial s_1}{\partial t_1} = s_2 x_1 s_1$$
$$\frac{\partial s_1}{\partial t_3} = 5 s_2$$
$$\frac{\partial s_1}{\partial t_2} = s_2 x_1$$
$$\frac{\partial s_2}{\partial t_1} = (-1) s_1^2 x_1$$
$$\frac{\partial s_2}{\partial t_3} = (-5) s_1$$
$$\frac{\partial s_2}{\partial t_2} = (-1) s_1 x_1$$